

Brain-only walkthrough: five-emotion NeuroEmo fMRI dataset

Detailed co-author speech notes and NeurIPS-style interpretation. This document excludes the separate anger/fear dataset and focuses only on the two uploaded brain-analysis HTML notebooks.

Item	Exact detail used in the brain-only analysis
Dataset	NeuroEmo: An fMRI Dataset for Emotion Recognition, OpenNeuro ds005700, version 1.2.0
Source	https://openneuro.org/datasets/ds005700/versions/1.2.0
Participants	40 healthy participants
Emotion labels	Afraid, Calm, Delighted, Depressed, Excited
Final analysed matrix	200 subject-emotion observations = 40 subjects x 5 emotions
Feature space	48 Harvard-Oxford cortical ROI features plus subject and emotion metadata
Main brain validation	LOSO decoding accuracy = 0.5600 vs chance = 0.2000; permutation $p < 0.001$
Core ambiguity validation	Centroid margin predicts cross-validated classifier uncertainty: Spearman $r = 0.6108$, $p = 7.78e-22$; AUC = 0.8124
Control	Feature-shuffled margin vs uncertainty collapses to $r = 0.0287$, $p = 0.687$

1. What this walkthrough is for

This is written as a detailed speech-style walkthrough for co-authors. It explains what has been done with the primary five-emotion brain dataset, what each analysis means, and how to describe the findings without leaving conceptual gaps. The emphasis is on clarity: what the vectors are, what high-dimensional plotting means, what centroids mean, what significant statistics mean, and why margin is the correct ambiguity metric rather than raw distance.

2. Dataset and scientific role

The primary brain dataset is NeuroEmo, an OpenNeuro BIDS-formatted fMRI dataset collected from 40 healthy participants during an emotion-elicitation task. The task uses emotionally evocative Indian movie clips selected to induce five emotional states: Afraid, Calm, Delighted, Depressed, and Excited. White-noise blocks are present in the original task, but they are excluded from the five-emotion geometry because they are a control condition rather than an emotion category.

The final object used for analysis is not raw fMRI images. It is a subject-by-emotion feature matrix. Each row corresponds to one participant under one emotion condition. Since there are 40 participants and 5 retained emotion labels, the final analysed representation contains 200 rows.

Co-author wording: The brain data provide a biological representation space. Each sample is one subject under one emotion, represented as regional activity across 48 cortical ROIs.

3. What the 48-dimensional space means

Each sample is a 48-dimensional vector because activity was extracted from 48 Harvard-Oxford cortical regions of interest. A dimension is not an emotion. It is one atlas-defined cortical region. Therefore a point in this space represents a distributed macro-scale brain activity pattern. The point does not say one region equals one emotion; it captures the joint pattern across regions.

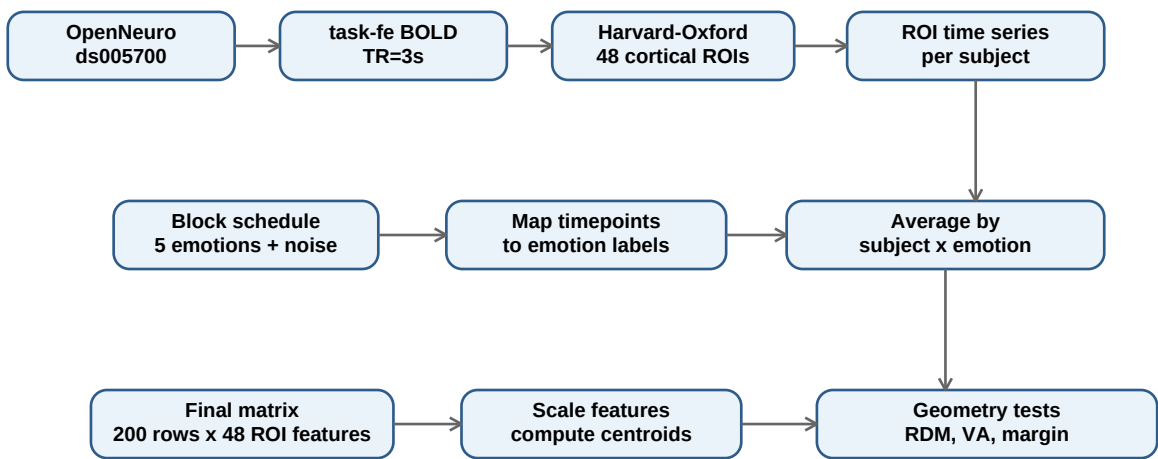
When we say we analyse the data in 48D space, we mean that every sample has 48 coordinates. We cannot visually inspect 48 dimensions directly, so 2D plots such as PCA are visual summaries. The actual centroid distances, margins, and RDMS are computed in the high-dimensional feature space unless explicitly stated

otherwise. This distinction matters because the visual plot is not the full statistical object.

If someone asks what plotting in high dimensions means: we are not drawing a literal 48-axis graph; we compute mathematical relationships in the 48-dimensional coordinate system, then optionally project them into two dimensions for interpretation.

4. Processing pipeline from raw fMRI to geometry

Participants watched emotion-inducing movie clips in a block design. Emotional and white-noise control blocks were temporally mapped to the fMRI BOLD signal. The functional scans were converted into ROI time series using Nilearn and the Harvard-Oxford cortical atlas. Each target emotion appeared twice in the task sequence. For each participant, timepoints belonging to the same emotion were averaged, yielding one stable ROI vector per participant per emotion.



This averaging step reduces block-level noise and produces a subject-level representation suitable for centroid geometry. The trade-off is that trial-level variability is compressed. For the current paper, this is acceptable because the goal is to establish a stable five-class neural geometry for cross-system comparison.

5. Scaling and why it is necessary

Before geometric analysis, ROI features were standardised. This prevents high-variance ROIs from dominating the Euclidean distance structure. Without scaling, geometry might reflect regional signal scale rather than multivariate emotion structure. After scaling, each ROI contributes on a comparable scale, making centroid distances and margins more interpretable.

6. Centroids and what they mean

An emotion centroid is the mean vector of all samples belonging to that emotion. For example, the Afraid centroid is the average 48D ROI pattern across all subjects under the Afraid condition. Centroids are statistical prototypes. They should not be described as pure emotions or literal neural states; they are average patterns around which subject-specific responses are distributed.

Centroid for emotion e : $c_e = \text{mean}\{x_i : y_i = e\}$

Pairwise centroid distance	Value
Afraid - Calm	1.1097
Afraid - Delighted	1.1826
Afraid - Depressed	1.2997

Afraid - Excited	0.9634
Calm - Delighted	1.3902
Calm - Depressed	1.0423
Calm - Excited	1.5073
Delighted - Depressed	1.0419
Delighted - Excited	1.1244
Depressed - Excited	1.4554

7. Decoding validation: showing the space contains emotion signal

Before interpreting geometry, the analysis validates that emotion information is actually present in the ROI space. This was done using leave-one-subject-out cross-validation. In each fold, all five samples from one subject are held out and the model is trained on the remaining subjects. This tests whether the emotional structure generalises to unseen participants.

Statistic	Result	Meaning
LOSO accuracy	0.5600	The classifier correctly predicts 56% of held-out subject-emotion samples.
Chance level	0.2000	With five classes, random guessing is 20%.
Balanced accuracy	0.5600	The result is not driven by class imbalance.
Accuracy 95% CI	[0.4900, 0.6300]	Bootstrap interval around the observed accuracy.
Balanced accuracy 95% CI	[0.4906, 0.6280]	Bootstrap interval around balanced accuracy.
Permutation null mean	0.1978	Expected accuracy under shuffled labels.
Permutation null SD	0.0334	Spread of shuffled-label accuracies.
Permutation p-value	$p < 0.001$	The observed decoding is significantly above the shuffled-label null.

The important interpretation is not that the brain data are perfectly separable. The correct interpretation is that the 48D ROI space contains real emotion-related structure, but that structure is overlapping and moderate rather than cleanly categorical. This makes it ideal for studying graded geometry and ambiguity.

8. Why geometry is different from classification

Classification asks whether we can predict the label. Geometry asks how points are arranged relative to prototypes and competitors. This matters because a sample can be correctly classified but still lie close to another emotion centroid, or it can be far from its own centroid but also far from every other centroid. Therefore ambiguity cannot be reduced to accuracy alone.

9. Distance alone is weak: why the naive assumption fails

A natural starting assumption is that points farther away from their correct centroid should be more ambiguous. We tested this directly by correlating distance to the correct centroid with classifier uncertainty, where uncertainty is defined as 1 minus the cross-validated probability assigned to the correct class.

The result was weak: Spearman $r = 0.1509$ with $p = 3.29e-02$; Pearson $r = 0.1471$ with $p = 3.76e-02$; AUC for detecting high-uncertainty samples = 0.5627. This means raw distance is statistically non-zero but practically weak and close to chance as a predictive measure.

Interpretation: being far from your own centroid does not necessarily mean you are ambiguous. A point can be far from its own prototype and still be farther from every other prototype. Ambiguity is not radial; it is relational.

10. Margin: the correct ambiguity mechanism

The stronger quantity is centroid margin. In the final analysis, margin is defined as the distance to the correct centroid minus the distance to the nearest competing centroid.

$$\text{margin}_i = d_{\text{correct}_i} - d_{\text{nearest_other}_i}$$

This metric measures competition. A large negative margin means the point is clearly closer to the correct centroid than to the nearest alternative. A margin near zero means the point lies close to a boundary between two emotional prototypes. A positive margin means the nearest competing centroid is closer than the labelled centroid.

Metric	Statistic	Interpretation
Margin vs uncertainty	Spearman $r = 0.6108$, $p = 7.78e-22$	Strong monotonic relationship.
Margin vs uncertainty	Pearson $r = 0.6351$	Strong linear association.
Margin AUC	0.8124	Strong ability to distinguish high- vs low-uncertainty samples.
Distance AUC	0.5627	Distance alone is close to chance.
Feature-shuffled margin	$r = 0.0287$, $p = 0.687$	The relationship disappears when multivariate structure is destroyed.

This is the core ambiguity result. We are no longer assuming that distance defines ambiguity. We show empirically that relative centroid competition predicts independent classifier uncertainty. This is the reviewer-safe version of the claim.

11. Why the shuffle control matters

The feature-shuffle control destroys the coordinated multivariate structure by shuffling each feature across samples. This preserves the distribution of each ROI feature, but breaks the relationship between features within individual samples. If the margin-uncertainty relationship were a by-product of random feature distributions, it would remain after shuffling. Instead, it collapses to $r = 0.0287$ with $p = 0.687$.

This supports the claim that the margin-uncertainty relationship is non-random and depends on the actual multivariate organisation of the neural representation.

Short version: randomness can produce distributions, but it does not produce a stable, reproducible relationship between centroid margin and uncertainty that disappears under randomisation.

12. Density and overlap: how they fit in

Density decay describes how points are distributed around their own centroids. It is useful for representational packing: whether samples are tightly concentrated or broadly dispersed. However, density is not the same as ambiguity. Density asks where points are. Margin asks whether a point is competing between prototypes.

Overlap decay is closer to the ambiguity claim because it examines whether points at different distances are more likely to have another centroid closer than their labelled centroid. In the paper, density should be presented as supporting evidence, while margin versus uncertainty should be the main ambiguity validation.

13. Valence-arousal alignment: global brain organisation

The valence-arousal analysis is separate from the margin result. It asks what psychological axis explains the largest differences between emotion centroids. PCA on the five brain centroids shows that PC1 explains 46.4% of centroid variance and aligns strongly with arousal.

VA statistic	Result	Meaning
PC1 explained variance	0.4642	PC1 captures 46.4% of centroid variance.
PC2 explained variance	0.3243	PC2 captures 32.4% of centroid variance.
PC1 vs Valence	$r = 0.255$, $p = 0.6789$	No meaningful PC1-valence alignment.
PC1 vs Arousal	$r = 0.965$, $p = 0.0078$	Dominant brain axis is strongly arousal-aligned.
PC2 vs Valence	$r = -0.064$, $p = 0.9185$	No meaningful PC2-valence alignment.
PC2 vs Arousal	$r = 0.039$, $p = 0.9504$	No meaningful PC2-arousal alignment.

Interpretation: the dominant global axis of the brain's five-emotion geometry is arousal rather than valence. This should not be overclaimed because it is based on five centroids, but it is a strong and interpretable effect. It is especially useful for explaining later why brain and LLM global geometries may differ even if local ambiguity behaviour can be compared.

14. What each significant value means

Finding	Statistic	Meaning
Emotion signal exists	LOSO 0.56 vs chance 0.20; $p < 0.001$	The 48D ROI patterns contain decodable emotion information.
Raw distance is weak	Spearman $r = 0.1509$; AUC = 0.5627	Distance alone is not a strong ambiguity measure.
Margin is strong	Spearman $r = 0.6108$; AUC = 0.8124	Relative prototype competition predicts uncertainty.
Not random	Shuffled $r = 0.0287$, $p = 0.687$	The relationship disappears after randomising feature structure.
Brain global axis	PC1-arousal $r = 0.965$, $p = 0.0078$	The dominant centroid axis is arousal-aligned.

15. Full co-author speech walkthrough

"For the brain side, we are using the NeuroEmo five-emotion fMRI dataset as the primary biological anchor. The final data are not raw fMRI images. They are a subject-by-emotion representation: 40 subjects, five emotion conditions, and 48 Harvard-Oxford cortical ROI features per sample, giving 200 brain vectors. Each vector is a distributed pattern of regional activity, not a single-region activation or a subjective confidence score."

"We first validate that this neural space contains emotion-related information. Leave-one-subject-out decoding gives 56% accuracy across five classes compared with 20% chance, and the permutation test is below 0.001. So emotion labels are decodable from the distributed ROI patterns, but the moderate accuracy tells us the brain space is overlapping rather than cleanly separated."

"We then compute emotion centroids in the scaled 48D space. These centroids are statistical prototypes: the average ROI pattern for each emotion. The point of the geometry is to ask how each subject-emotion vector sits relative to its own prototype and competing prototypes."

"A key methodological refinement is that raw distance to the correct centroid is not enough. It weakly predicts uncertainty, with r around 0.15 and AUC around 0.56. So we cannot say far from centroid equals ambiguous. That assumption is too simplistic."

"The correct measure is centroid margin: distance to the correct centroid minus distance to the nearest competing centroid. This captures class competition. Margin strongly predicts classifier uncertainty, with Spearman $r = 0.61$ and $AUC = 0.81$. When we shuffle features, the relationship disappears. That means ambiguity is not random and not just a distance effect; it emerges from relative positioning between competing emotion prototypes."

"Separately, the global valence-arousal analysis shows that the dominant brain centroid axis aligns with arousal, not valence. PC1 explains about 46% of centroid variance and correlates with arousal at $r = 0.965$. So the brain data are globally arousal-dominant, while the margin result tells us about local boundary competition."

"The final brain-only conclusion is that the NeuroEmo five-emotion ROI space is decodable, structured, and geometrically interpretable. The strongest result is that uncertainty is governed by centroid competition rather than raw distance. This gives us a robust biological basis for later comparison with LLM embedding spaces."

16. Recommended paper phrasing

Use: "Ambiguity is operationalised as centroid competition and validated against cross-validated classifier uncertainty."

Use: "Absolute distance is insufficient; relative margin captures the relevant decision geometry."

Use: "The brain's dominant global centroid axis is arousal-aligned, while local ambiguity is governed by prototype competition."

Avoid: "Centroids are pure emotions", "distance equals ambiguity", "the brain has confidence scores", or "the geometry proves a universal law".

17. Final conclusion

The brain-only analysis now supports a coherent, defensible methodological story. The dataset is well-defined; the ROI feature construction is biologically interpretable; the five-class neural representation is significantly decodable; centroid geometry provides statistical prototypes; and ambiguity is validated as a relative margin effect rather than assumed from raw distance. This is the strongest form of the brain-side evidence for the paper.